Master's Thesis Summary
Centro de Investigación en Matemáticas (CIMAT)

Non-lifting of the Mapping Class Group to the Diffeomorphism Group of a Surface with Marked Points

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The mapping class group of a closed orientable surface S_g of genus g with a finite set of marked points $\mathbf{z} = \{z_1, \dots, z_k\}$ is defined as the quotient group

$$\mathrm{Mcg}(S_g, \mathbf{z}) := \frac{\mathrm{Diff}^+(S_g, \mathbf{z})}{\mathrm{Diff}_0(S_g, \mathbf{z})},$$

where $\mathrm{Diff}^+(S_g, \mathbf{z})$ denotes the group of orientation-preserving diffeomorphisms that preserve the set $\mathbf{z}$, and $\mathrm{Diff}_0(S_g, \mathbf{z})$ is the path component of the identity.

In this context, the lifting problem of mapping class groups asks whether a subgroup $\Gamma \leq \mathrm{Mcg}(S_g, \mathbf{z})$ admits a lift to the diffeomorphism group that makes the following diagram commute.

$$\begin{array}{ccc} & & \mathrm{Diff}^+(S_g, \mathbf{z}) \\ & \overset{\sigma}{\nearrow} & \big\downarrow{\scriptstyle [\cdot]} \\ \Gamma & \hookrightarrow & \mathrm{Mcg}(S_g, \mathbf{z}). \end{array}$$

Following the approach of Bestvina, Church and Souto [BCS13], in this thesis we provide a detailed proof of the following theorem.

Theorem 1. Let S_g be a closed orientable surface of genus $g \geq 2$, and let $\mathbf{z} = \{z_1, \dots, z_k\}$ be a set with $k \geq 1$ marked points. Then no finite-index subgroup of $\mathrm{Mcg}(S_g, \mathbf{z})$ admits a lift to the diffeomorphism group.

The previous theorem is an algebraic consequence of the one-marked-point case. In this setting, there is an embedding $\pi_1(S_g, z) \hookrightarrow \mathrm{Mcg}(S_g, z)$, known as the *point-pushing* map. We therefore consider the following theorem.

Theorem 2. For a closed oriented surface S_g of genus $g \geq 2$ and a marked point $z \in S_g$, the point-pushing map does not admit a lift that makes the diagram

$$\begin{array}{ccc} & & \mathrm{Diff}^+(S_g, z) \\ & \overset{\sigma}{\nearrow} & \big\downarrow{\scriptstyle [\cdot]} \\ \pi_1(S_g, z) & \xhookrightarrow{\text{push}} & \mathrm{Mcg}(S_g, z) \end{array}$$

commute. In particular, $\mathrm{Mcg}(S_g, z)$ does not admit such a lift.

To find an obstruction to such a lift, we study the Euler class of oriented circle bundles over S_g. For an oriented circle bundle $\xi = (\mathbb{S}^1 \hookrightarrow E \to S_g)$ the Euler class $e(\xi) \in H^2(S_g, \mathbb{Z}) \cong \mathbb{Z}$ is a characteristic class whose evaluation on the fundamental class $[S_g]$ gives the Euler number

$$e(\xi)[S_g] \in \mathbb{Z}.$$

The classification of oriented circle bundles [Mor01, Theorem 6.22] and Milnor's formula [Mil58] allow us to study flat circle bundles, i.e., those constructed by the usual action of the fundamental group on the universal cover, together with a prescribed representation $\rho : \pi_1(S_g, z) \to \mathrm{Homeo}^+(\mathbb{S}^1)$. This provides a diagonal action $\pi_1(S_g, z) \curvearrowright \widetilde{S}_g \times \mathbb{S}^1$ and an associated circle bundle

$$E_\rho := \left(\widetilde{S}_g \times \mathbb{S}^1\right) \Big/ \pi_1(S_g, z) \to S_g,$$

that we call a flat circle bundle. The Milnor-Wood inequality can be stated as follows.

Theorem 3 (Milnor-Wood inequality, [Woo71])**.** Let S_g be a closed orientable surface with genus $g \geq 0$, and let $\rho : \pi_1(S_g, z) \to \mathrm{Homeo}^+(\mathbb{S}^1)$ be a representation. The Euler number of the associated flat circle bundle $E_\rho \to S_g$ satisfies the following inequality

$$|e(E_\rho)[S_g]| \leq \begin{cases} 2g-2 = -\chi(S_g) & \text{if } g \geq 1, \\ 0 & \text{if } g = 0. \end{cases}$$

Moreover, if $\rho : \pi_1(S_g, z) \to \mathrm{Homeo}^+(\mathbb{S}^1)$ commutes with the antipodal map $a : \mathbb{S}^1 \to \mathbb{S}^1$, then the following sharper bound holds

$$|e(E_\rho)[S_g]| \leq \begin{cases} g-1 = -\frac{1}{2}\chi(S_g) & \text{if } g \geq 1 \\ 0 & \text{if } g = 0. \end{cases}$$

In this thesis, written in Spanish, we provide detailed proofs of Milnor's formula and the Milnor-Wood inequality. We then combine these results to prove Theorem 2 by contradiction. Assuming that such a lift exists, we construct an annulus bundle $\mathbb{S}^1 \times [0,1] \hookrightarrow E \to S_g$ whose boundary consists of two flat circle bundles. The first is the unit tangent bundle, whose Euler number is

$$e(UTS_g)[S_g] = 2 - 2g,$$

while the latter is a linear circle bundle, associated to a representation $\pi_1(S_g, z) \to \mathrm{GL}_2^+(\mathbb{R}) \to \mathrm{Homeo}^+(\mathbb{S}^1)$ by normalization, whose Euler number is bounded in absolute value by $g-1$ by Theorem 3. The differentiability assumption is essential for constructing an action by linear transformations that gives rise to the latter circle bundle.

The fact that both circle bundles arise as boundary components of the same annulus bundle implies that their Euler numbers must be equal, yielding a contradiction.

My contribution

The precise contribution of my work is the reasoning of why the euler numbers of the boundary circle bundles should be equal. <++>

Additional contents

References

[BCS13] Mladen Bestvina, Thomas Church, and Juan Souto. "Some groups of mapping classes not realized by diffeomorphisms". In: *Comment. Math. Helv.* 88.1 (2013), pp. 205–220. ISSN: 0010-2571,1420-8946. DOI: 10.4171/CMH/283. URL: https://doi.org/10.4171/CMH/283.

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[Mor01] Shigeyuki Morita. *Geometry of Differential Forms*. Vol. 201. Translations of Mathematical Monographs. Translated from the two-volume Japanese original (1997, 1998) by Teruko Nagase and Katsumi Nomizu, Iwanami Series in Modern Mathematics. American Mathematical Society, Providence, RI, 2001, pp. xxiv+321. ISBN: 0-8218-1045-6. DOI: 10.1090/mmono/201. URL: https://doi.org/10.1090/mmono/201.

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Obstructions to the realization of subgroups of mapping classes by diffeomorphisms

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Introduction

Given a compact orientable surface S_g and a finite set $\mathbf{z} = \{z_1, \dots, z_k\}$ we obtain the punctured surface $S_g^k := S_g \setminus \mathbf{z}$ or the marked surface $(S_g, \mathbf{z})$.



Their **mapping class groups** are isomorphic

$$\mathrm{Mcg}(S_g^k) := \frac{\mathrm{Diff}^+(S_g^k)}{\mathrm{Diff}_0(S_g^k)} \cong \mathrm{Mcg}(S_g, \mathbf{z}) := \frac{\mathrm{Diff}^+(S_g, \mathbf{z})}{\mathrm{Diff}_0(S_g, \mathbf{z})}$$

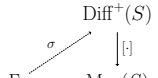
where Diff_0 is the path-connected component (normal subgroup) of the identity.

The group $\mathrm{Diff}^+(S)$ **acts naturally** on S. The question is:

Is there a natural action of $\mathrm{Mcg}(S)$ on the surface S as well?

The realization problem

Which subgroups $\Gamma \leq \mathrm{Mcg}(S)$ admit a **lift** to the diffeomorphism group $\mathrm{Diff}^+(S)$. That is a group homomorphism $\sigma : \Gamma \to \mathrm{Diff}^+(S)$ that makes the diagram



commute?

Answers for closed surfaces

Positive answers. Subgroups of $\mathrm{Mcg}(S_g)$ that have been proven to admit a lift to $\mathrm{Diff}^+(S_g)$ are all the following:

free subgroups, *finite cyclic subgroups* (Nielsen, 1943), *finite subgroups* (Kerckhoff, 1983), and *abelian subgroups* (Birman-Lubotzky-McCarthy, 1983).

Negative answers. For genus $g \geq 5$, Morita proved that $\mathrm{Mcg}(S_g)$ does not admit such a lift [1].

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Non-lifting for punctured surfaces.

Theorem. (Bestvina, Church and Souto, [2].) Given a genus $g \geq 2$ compact orientable surface S_g and a set of $k \geq 1$ marked points $\mathbf{z} = \{z_1, \dots, z_k\}$

- finite index subgroups of the *point-pushing subgroup* $\mathbf{push}(\pi_1(S_g, z_1)) \leq \mathrm{Mcg}(S_g, z_1)$ do not admit a lift to $\mathrm{Diff}^+(S_g, z_1)$;
- finite index subgroups of $\mathrm{Mcg}(S_g, \mathbf{z})$ do not admit a lift to $\mathrm{Diff}^+(S_g, \mathbf{z})$.

Point-pushing map

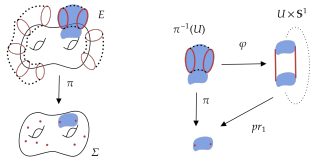
There is a monomorphism $\mathbf{push} : \pi_1(S_g, z) \hookrightarrow \mathrm{Mcg}(S_g, z)$ described by the following figure.



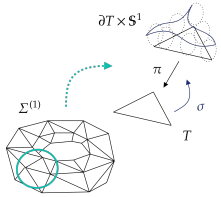
Its image $\mathbf{push}(\pi_1(S_g, z)) \cong \pi_1(S_g, z)$ is known as the *point-pushing subgroup*.

Euler class for $\mathbb{S}^1$-bundles

A **circle bundle** (or $\mathbb{S}^1$**-bundle**) over a surface is given by a surjection $\pi : E \to S_g$ whose fibers are homeomorphic to $\mathbb{S}^1$ and which admits local trivializations.

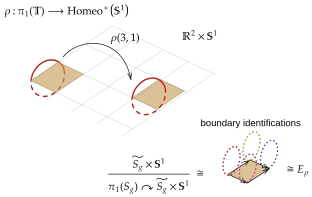


The **Euler class** of an **orientable** $\mathbb{S}^1$-bundle $E \to S_g$ is a characteristic class $e(E) = [c] \in H^2(S_g, \mathbb{Z})$ given by a cocycle, such that $c(T) = \deg(\text{map in the figure})$.



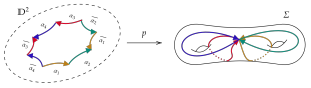
Flat circle bundles

Given a representation $\rho : \pi_1(S_g, z) \to \mathrm{Homeo}^+(\mathbb{S}^1)$, we produce a **diagonal** action $\pi_1(S_g, z) \curvearrowright \widetilde{S}_g \times \mathbb{S}^1$, whose quotient is an $\mathbb{S}^1$-bundle E_ρ over S_g.



In the first coordinate of $\widetilde{S}_g \times \mathbb{S}^1$ the action is described by **deck transformations** and in the second by ρ.

Remark. When $g \geq 2$ the universal cover of S_g admits a hyperbolic geometry, so the deck transformations can be realized by hyperbolic isometries.



Milnor-Wood inequality

Milnor gave a formula to compute the Euler class for flat bundles [3]. Later in [4], Wood proposed the upper bound for the Euler number of the flat bundle associated to $\rho : \pi_1(S_g, z) \to \mathrm{Homeo}^+(\mathbb{S}^1)$

$$|e(E_\rho)[S_g]| \leq \begin{cases} 2g - 2 = -\chi(S_g) & \text{if } g \geq 1, \\ 0 & \text{if } g = 0. \end{cases}$$

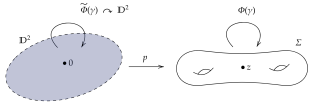
Moreover, if every element of $\rho(\pi_1(S_g, z))$ commutes with the antipodal map $\mathbf{a}(x) = -x$, then we get a better bound

$$|e(E_\rho)[S_g]| \leq \begin{cases} g - 1 = -\frac{1}{2}\chi(S_g) & \text{if } g \geq 1, \\ 0 & \text{if } g = 0. \end{cases}$$

Proof by contradiction

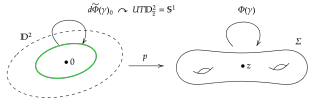
Suppose that **the point-pushing map lifts** to some homomorphism $\phi : \pi_1(S_g, z) \to \mathrm{Diff}^+(S_g, z)$.

(A) After lifting to the universal cover we obtain a representation $\widetilde{\phi} : \pi_1(S_g, z) \to \mathrm{Diff}^+(\mathbb{D}^2, 0)$.

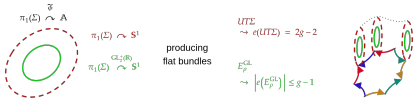


This action extends to $\partial_\infty \mathbb{D}^2$, where it agrees with the action of $\pi_1(S_g, z)$ by **deck transformations**.

(B) There is a **linear action** on the unit tangent circle given by $\pi_1(S_g, z) \xrightarrow{d\widetilde{\phi}_0} \mathrm{GL}_2^+(\mathbb{R}^2) \xrightarrow{\text{normalization}} \mathrm{Homeo}^+(\mathbb{S}^1)$.



(C) We combine both actions into a compactification of $\overline{\mathbb{D}}^2 \setminus \{0\}$ to the annulus $\mathbb{A}$. This produces an $\mathbb{A}$-bundle over S_g.



A retraction of the $\mathbb{A}$-bundle to each $\mathbb{S}^1$-bundle yields $e(E_1) = e(E_2)$. This contradicts the Milnor-Wood inequality

$$|e(E_1)[S_g]| \leq g - 1 < 2g - 2 = e(E_2)[S_g].$$

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